Warn clients two hours before a flight goes wrong, and turn a bad surprise into a managed one

1 in 4

flights were late or cancelled, Jan-Sep 2023 (25.7% of 330,883) [01]

2 hours

of notice: time to warn clients and prepare alternatives

2 alerts

tested once on Oct-Dec flights the model never saw [07]

The ask: approve a pilot of two alerts, with a person checking every one.

Data: nycflights23, all flights from JFK, LGA and EWR in 2023 (435,352 flights). Mirna Embaby

Sources: notebooks 01, 07

Three questions, and two traps in the data that we had to fix

The three questions

Q1 Which flights will be late or cancelled? **(Alert 1)**

Q2 Who is most and least reliable, and how much is weather? **(findings)**

Q3 Which flights will be cancelled? **(Alert 2)**

435,352 flights, 5 files, no row removed. [01]

The two traps

1. Weather is in UTC, flights are in New York time.

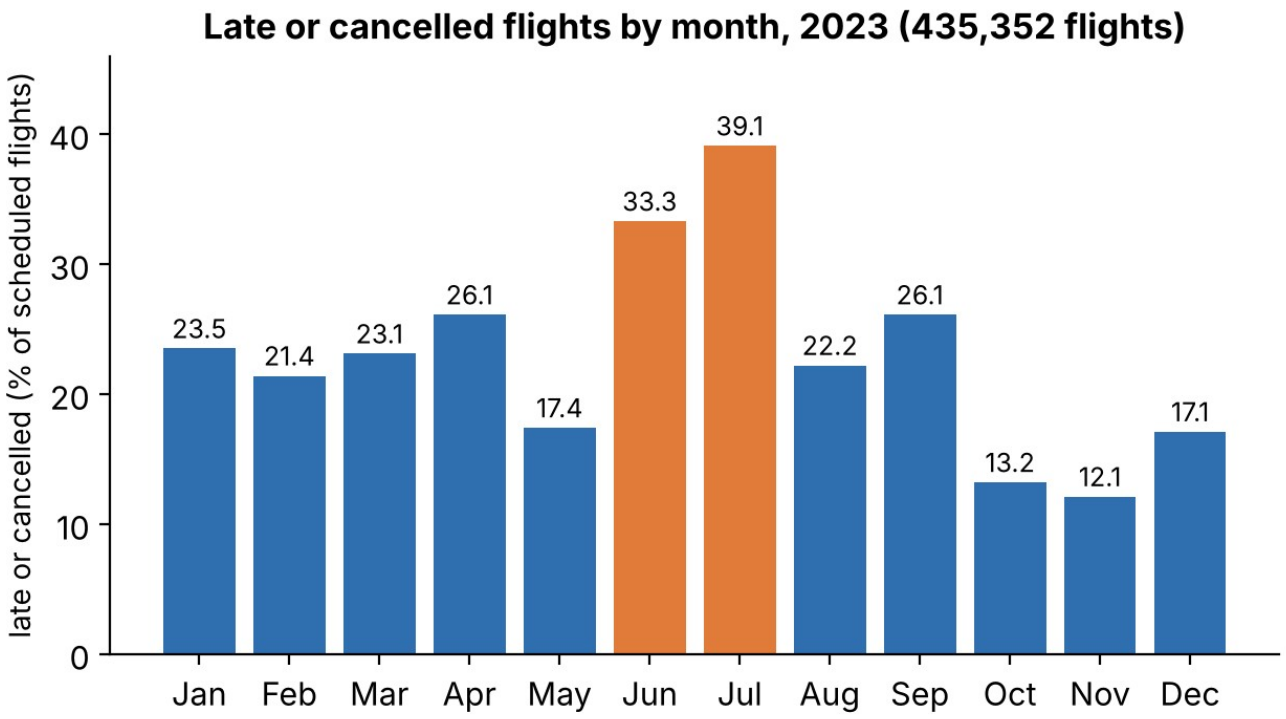
We converted first. A literal join would change the score by only −0.0039. [06]

2. No tail number = cancelled (1,913 of 1,913 flights). [01]

We fill plane data with a random real aircraft of the same airline: leak +0.012, not +0.206. [05]

Sources: notebooks 01, 05, 06

1 flight in 4 is late or cancelled, and July is the worst month



Orange = above 30%. About 36,000 flights a month. [04]

June and July were not wetter or foggier: a few bad days drive them, and the cause is **not explained by the available data**. [04]

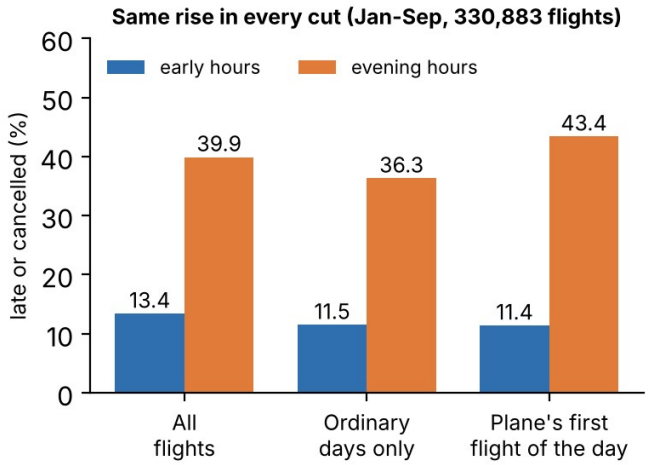
Metric, whole year [04]	Value	Out of
M1 On time (delay ≤ 15 min)	79.03%	departed flights
M2 Mean delay	13.84 min	departed flights
M3 Cancelled	2.47%	scheduled flights
M4 More than 1 h late	8.41%	departed flights
M5 Late or cancelled	22.91%	scheduled flights

Today a client finds out at the airport. We want them to know two hours earlier.

Sources: notebook 04 (whole year, 435,352 flights)

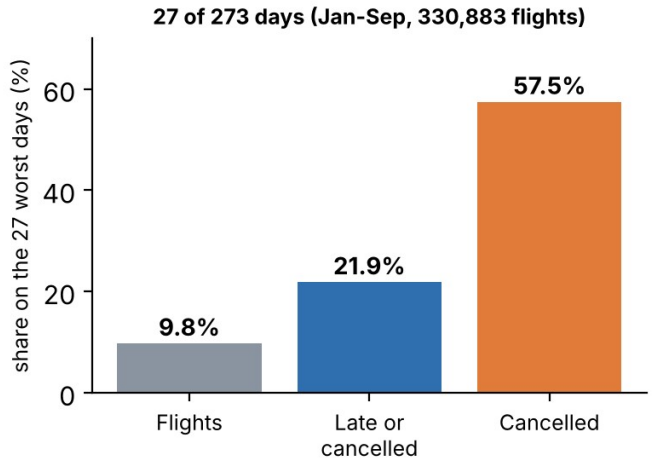
Three findings: risk builds through the day, a few days do the damage, and airport state beats weather

1. Risk builds through the day



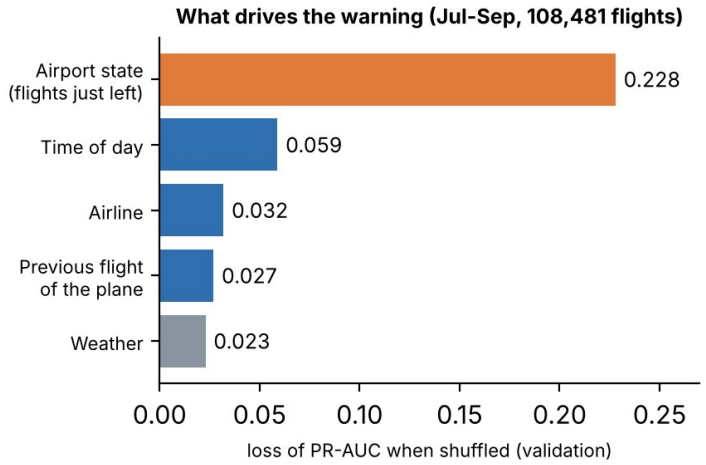
13.4% → **39.9%** late or cancelled, early vs evening (gap 26.5 points, interval 24.2 to 28.9). [02]

2. A few bad days do the damage



27 days out of 273 hold **57.5%** of all cancellations. [02]

3. Airport state beats weather



Airport state **0.228** against weather **0.023**. Rain adds 14.5 points. [06]

Action: Warn earlier for evening flights.

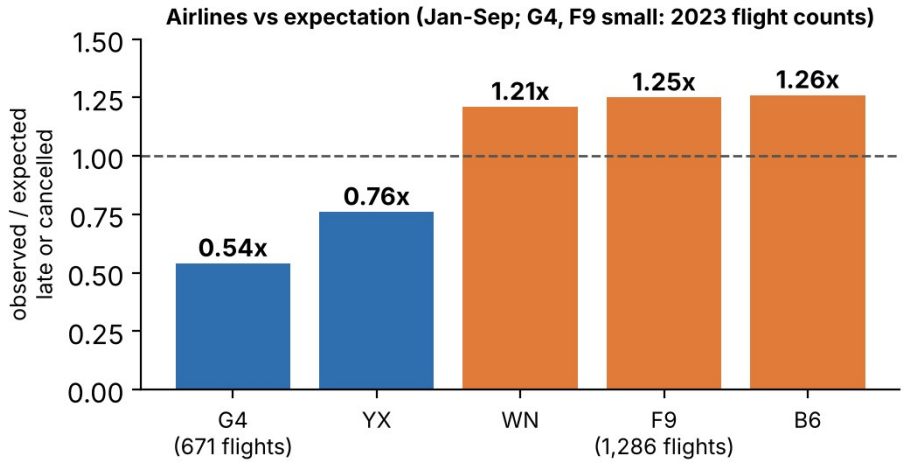
Action: Add a bad-day rule.

Action: Watch the airport live.

Sources: notebook 02 (Jan-Sep, day-bootstrap intervals), notebook 06 (importance)

Airlines differ on bad days, and the weekday pattern was an accident of the calendar

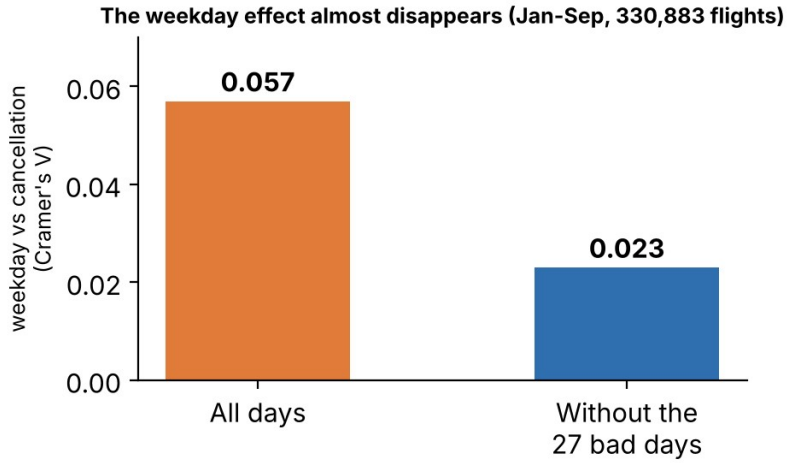
4. Airlines



On bad days **YX cancels 29%** and 9E 27%, **Delta 9%**. On ordinary days every big airline cancels 1.0% to 1.7%. [02]

Action: rebooking ready first for YX and 9E.

5. A strange pattern, explained



Thursday 1.5% vs Sunday 4.7% cancelled looks like a weekday effect. It nearly vanishes without the 27 bad days (**0.057 → 0.023**). [02]

Action: do not plan by weekday.

Sources: notebook 02 (Jan-Sep)

Destinations fall into three groups, each with its own first action

Short-haul

57 airports · 101,287 flights (Jan-Sep) [03]

21.2% late

Most cancellations (4.3%)

Action: Have rebooking ready first.

Main long-haul

45 airports · 226,896 flights (Jan-Sep) [03]

27.8% late

Late 15% in the morning, 42% in the evening

Action: Warn earlier on evening flights.

Chronic delays

7 airports · 2,374 flights (Jan-Sep) [03]

53.1% late

Mean delay 41.5 minutes

Action: Always warn and check connections.

The groups are a guide, not a certainty (silhouette 0.322). [03]

Sources: notebook 03 (k-means, Jan-Sep)

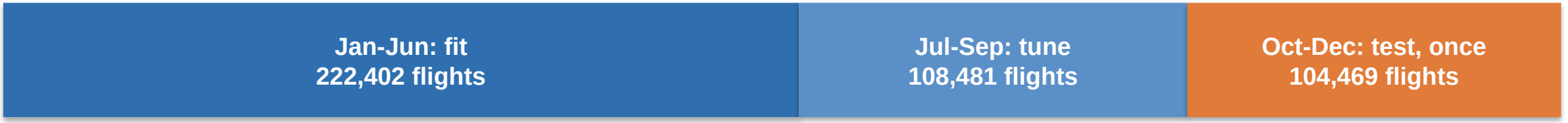
Each score uses only what is known two hours before, and is tested once on months it never saw

Known two hours before (used)
schedule · route · flights that already left · previous flight of the plane · weather · plane details [05]

T = 2 h before departure

After T (never used)
actual times · delays · cancellation flags [05]

4 automated leak tests, all pass [05]

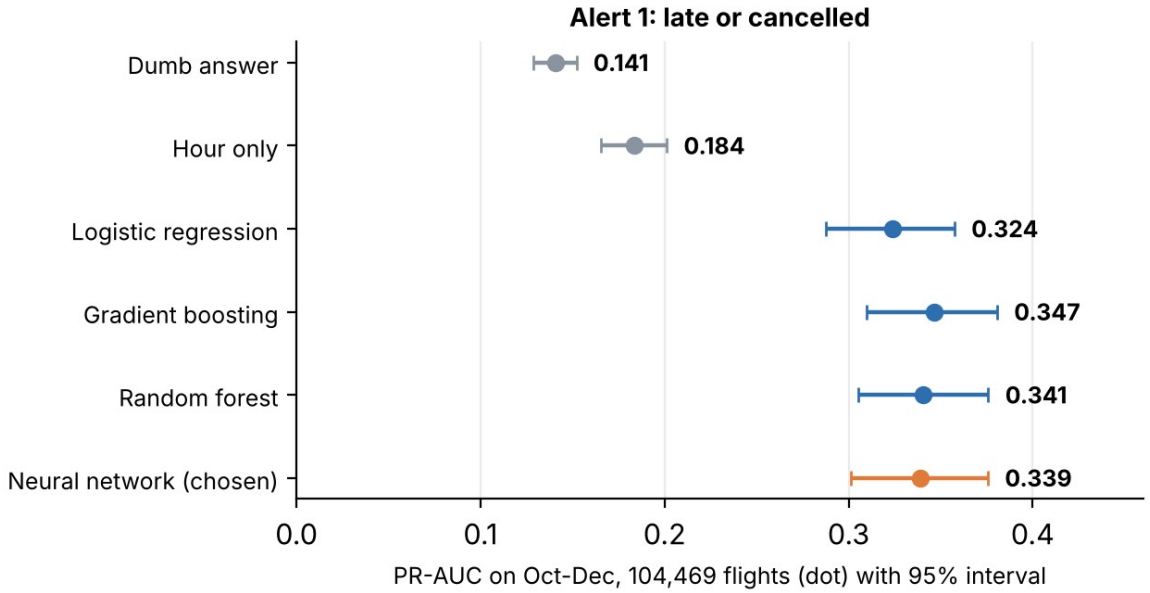


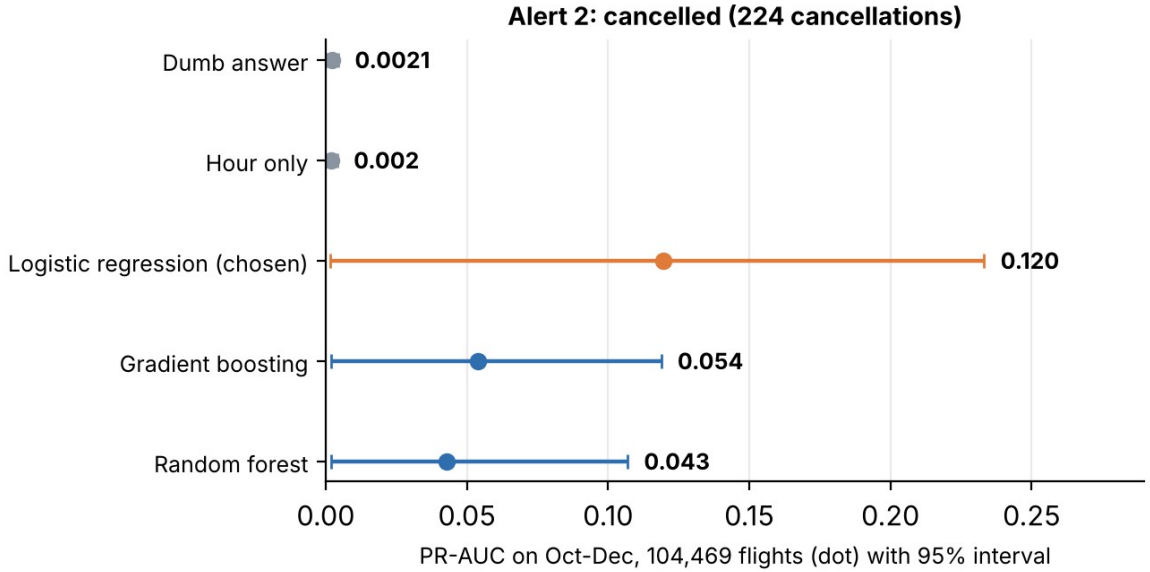
Final model refit on Jan-Sep (330,883 flights). [06]

Why not accuracy? "Nobody is late" is **85.9% accurate** and finds nothing. We compare every model with that dumb answer using PR-AUC: **0.1406** (Alert 1), **0.0021** (Alert 2). [06]

Sources: notebook 05 (leak tests), notebook 06 (protocol)

Both alerts beat the dumb answer; the models are close, and cancellation is uncertain





Alert 1

Neural network, best on validation. On test all four models are close. Without airport state: −0.058. [06]

Alert 2

Logistic regression. Very wide interval: 224 cancellations, 115 on one day. [06]

Weather choices

Strict weather −0.0016 and literal join −0.0039: both inside the interval. [06]

Sources: notebook 06 (final tables, ablation rows)

Alert 1: about 130 flights flagged a day, and more than one in three really goes wrong

130 a day

flights flagged (11,974 in 3 months; cut-off 0.597) [07]

37.4%

of flags really late or cancelled (interval 34.2 to 40.6); random: 14.1% [07]

30.5%

of problem flights caught (interval 25.4 to 35.1); 2.7x random [07]

0.339

PR-AUC (interval 0.301 to 0.376) vs 0.141 for the dumb answer [06]

What to do

Message the client two hours before and check connections. A person confirms the flight status first.

Watch out

Weak in the morning (5.8% of problem flights caught) and at Newark (20.5%). [06]

Re-check when

Weekly recall is below 14% two weeks running (weeks with 100+ cases: 14.3% to 51.7%). [07]

Question: more than 15 minutes late or cancelled, asked two hours before departure. **Model:** neural network trained on Jan-Sep 2023 flights. **Why recall counts:** a false alarm costs one message; a missed delay leaves a client unwarned.

Finding: airport state two hours earlier is worth more than all weather columns (without it: −0.058 PR-AUC). [06]

Sources: notebook 07 (card 1, Oct-Dec test, 104,469 flights), notebook 06

Alert 2: about 5 flights a day to prepare rebooking for, catching 42% of cancellations

5.3 a day

flights flagged (484 in 3 months; cut-off 0.137) [07]

19.4%

of flags really cancelled (interval 0.6 to 25.9); random: 1 in 500 [07]

42%

of cancellations caught: 94 of 224 (interval 1.0 to 60.5); 91x random [07]

0.120

PR-AUC (interval 0.002 to 0.233) vs 0.0021 for the dumb answer [06]

What to do

Offer a rebooking before the airline cancels. Never auto-rebook.

Watch out

Missed all 39 Newark cancellations. 115 of 224 fell on one day (18 Dec). [07]

Re-check when

Recall is below 23% in a bad-day block (half of the 46.7% seen in validation). [07]

Question: will the flight be cancelled, asked two hours before departure. **Model:** logistic regression trained on Jan-Sep 2023 flights. **Why recall counts:** a false alarm costs one message; a missed cancellation leaves a client stranded.

Use it as a ranking: the top 1% of scores catches 47.8% of cancellations (48x random). [07]

Sources: notebook 07 (card 2, Oct-Dec test, 104,469 flights), notebook 06

The limits are named, and a weekly check tells us if the alerts stop working

What it cannot do

- **Mornings:** only 5.8% of problem flights caught. [06]
- **Newark:** 20.5% for Alert 1; none of the 39 cancellations for Alert 2. [06]
- **Fairness:** flags JetBlue most; Alaska, American and Endeavor less than their real risk. [06]
- **Drift:** late or cancelled fell from 25.7% to 14.1%. [01]
- **Weather** is observed, not forecast (strict version: −0.0016). [06]

Weekly check and stop rules

	Watch weekly	Stop when
Alert 1	share of problem flights caught	below **14%** two weeks running
Alert 2	cancellations caught (4-week blocks)	below **23%** in a bad-day block

Always: a named person confirms the flight status before any client contact. [07]

These rules rest on 14 weeks of data. They are **tentative**: review after the first quarter. [07]

Sources: notebooks 01, 06, 07

Built with AI and checked by hand: every number traces back to the code

How AI (Claude) helped

- Wrote and debugged code (memory crash, tie bug, divide-by-zero)
- Helped plan the decisions
- Helped word the explanations and slides

What I checked myself

- The 15 metric check values, asserted in code [04]
- 4 leak tests on every feature [05]
- Test months scored once
- All 7 notebooks re-run top to bottom: same numbers

Every number has a tag [01]-[07] and a line in the README ledger.

Data: nycflights23 (C. Ismay, S. P. Couch, H. Wickham, 2025, CC0).

Sources: README.md, notebooks 01-07

Three yeses and we start a careful pilot

1. **Approve Alert 1:** about 130 messages a day [07], a person confirms each.
2. **Approve Alert 2:** about 5 flights a day [07] for rebooking readiness, never automatic.
3. **Approve the weekly check** with one named owner.

To make it stronger

a real weather forecast

the time each cancellation became known

a bad-day trigger [02]

What I need from you

how many messages a day the team can send

who confirms the flight status

how long the pilot runs

Sources: notebooks 02, 07